Project Analysis: ServiceNow Problem Ticket Analysis Dashboard (Excel)

Prepared by: Jasmine | **Dataset:** servicenow_m2_accelerated.csv | **Tool:** Microsoft Excel

All figures in this document are calculated from the 2,000-ticket CSV file. No numbers are estimated.

1. Project Name

ServiceNow Problem Ticket Analysis Dashboard (Excel)

2. Project Summary

This project turns a raw export of 2,000 ServiceNow problem tickets into an Excel workbook with cleaned data, KPI calculations, PivotTables, charts and an interactive dashboard. It shows how large the problem backlog is, how urgent and how old the open tickets are, and which teams carry the most work.

3. What the Project Does

1. Imports the CSV into Excel with Power Query and loads it as a table (tblProblems).
2. Cleans the data: sets Created Date to a date type and Age (Days) to a whole number, trims text, and checks for duplicate ticket IDs (none found).
3. Adds helper columns: **Status Group** (Active or Completed), **Age Bucket** (0–15, 16–30, 31–60, 61–90 days) and **Month**.
4. Calculates KPIs with COUNTA, COUNTIFS and AVERAGEIFS.
5. Builds PivotTables and PivotCharts for State, Priority, Team, Age Bucket, Month and Top Causes.
6. Combines them into a dashboard with KPI cards and slicers (Assigned Team, Priority, State, Month).
7. Documents findings and recommendations.

4. Business Issue the Project Solves

ServiceNow problem tickets are exported as a flat file of 2,000 rows. In that form, managers cannot easily answer:

- How many problems are still unresolved?
- Which unresolved problems are Critical or High priority?
- How old are the open problems?
- Which teams hold the most work?
- Which causes keep repeating?

The dashboard answers each of these in one view, and replaces manual counting with formulas and PivotTables that refresh when a new export is pasted in.

5. Dataset

Item	Detail
Records	2,000 problem tickets
Columns	9 (Problem Ticket, Problem Statement, Description, Cause, State, Priority, Assigned Team, Created Date, Age (Days))
Date range	30 July 2025 to 27 October 2025
Missing values	None
Duplicate ticket IDs	None
Teams	6
States	5 (Open, In Progress, Awaiting Info, Resolved, Closed)
Priorities	4 (Critical, High, Medium, Low)

6. Key Metrics

6.1 Headline KPIs

KPI	Value
Total tickets	2,000
Active tickets (Open, In Progress, Awaiting Info)	1,399 (69.95%)
Completed tickets (Resolved, Closed)	601 (30.05%)
Critical tickets	301
Critical tickets still active	213 (70.8% of Critical)
High tickets still active	340
Critical and High tickets still active	553
Average age of active tickets	45.5 days
Active tickets older than 30 days	929 (66.4% of active)
Active tickets older than 60 days	457 (32.7% of active)
Awaiting Info tickets	311 (22.2% of active)

6.2 Tickets by State

State	Tickets
In Progress	554
Open	534
Resolved	401
Awaiting Info	311
Closed	200

6.3 Tickets by Priority

Priority	Tickets
Medium	807
High	496
Low	396
Critical	301

6.4 Active Tickets by Age Bucket

Age bucket	Active tickets
0–15 days	225
16–30 days	245
31–60 days	472
61–90 days	457

6.5 Workload by Team

Team	Total tickets	Active tickets	Critical active
CMDB Team	478	323	41
Database Operations	428	293	55
Backup & Recovery	412	288	41
Infrastructure	240	171	25
Server Management	231	172	27
Network Operations	211	152	24

6.6 Tickets Created by Month

Month	Tickets
July 2025 (from 30 July only)	49
August 2025	675
September 2025	656
October 2025 (to 27 October)	620

6.7 Most Frequent Causes

Cause	Tickets
Resource allocation for web proxy requires optimization	78
Resource allocation for network monitoring requires optimization	75
Resource allocation for application gateway requires optimization	73
Resource allocation for Active Directory requires optimization	71
Resource allocation for backup verification requires optimization	70

7. Key Findings

1. **The backlog is large.** 1,399 of 2,000 tickets (69.95%) are still Open, In Progress or Awaiting Info. Only 30.05% are Resolved or Closed.
2. **Critical problems are mostly unresolved.** 213 of 301 Critical tickets (70.8%) are still active. Including High priority, 553 high-risk tickets are active.
3. **The backlog is ageing.** 929 active tickets (66.4%) are older than 30 days, and 457 are older than 60 days.

4. **Work is concentrated in three teams.** CMDB Team, Database Operations and Backup & Recovery hold 1,318 of 2,000 tickets (65.9%) and 904 of the 1,399 active tickets (64.6%).
5. **Database Operations has the most Critical active tickets.** It has 55 of the 213.
6. **Many tickets wait on information.** 311 tickets are Awaiting Info.
7. **Causes repeat.** The top five causes each appear 70 to 78 times, all following the same "resource allocation requires optimization" pattern across different services.

8. Recommendations

These are suggestions based on the findings, not results that have been measured.

- Review the 213 unresolved Critical tickets first and confirm an owner for each.
- Review the 457 active tickets older than 60 days.
- Check whether CMDB Team, Database Operations and Backup & Recovery have enough capacity for their active workload.
- Follow up on the 311 Awaiting Info tickets to find what information is missing.
- Look at the repeating "resource allocation" causes for a permanent fix instead of ticket-by-ticket handling.

9. Skills Demonstrated

Area	Skills
Data preparation	Data cleaning, data type conversion, duplicate checks, derived fields
Analysis	KPI design, backlog analysis, priority and ageing analysis, workload analysis, root-cause pattern analysis
Visualisation	Dashboard design, KPI cards, PivotCharts, slicers
Business analysis	BRD writing, requirement definition, insight and recommendation writing
Domain	ITSM, ServiceNow problem management

10. Technology Used

Tool	Use
Microsoft Excel	Workbook, tables, formulas, dashboard
Power Query	CSV import and cleaning
Excel Tables	Structured references (tblProblems)
Formulas	COUNTA, COUNTIFS, AVERAGEIFS, IF, OR, TEXT
PivotTables and PivotCharts	Summaries and charts
Slicers	Interactive filtering
Conditional formatting	Priority colour coding
CSV	Source data format

11. Data Limitations

- **Age (Days) is days since creation**, not time to resolve. Resolved and Closed tickets show an age too, so resolution time (MTTR) cannot be calculated.
- The file has **no resolved date, assignee or service field**, so SLA breaches and individual workload cannot be measured.
- The **Description** field contains generic text and was not used for the analysis.
- The data covers a single 90-day snapshot.

12. Deliverables

1. Business Requirements Document (BRD)
2. Excel workbook (Data, Calc, Pivots, Dashboard, Insights)
3. One-page dashboard exported as PDF
4. Project analysis (this document)

13. Possible Next Steps

- [ ] Add Resolved Date and Assignee to the export so that resolution time and SLA analysis become possible.
- [ ] Rebuild the same analysis in Power BI.